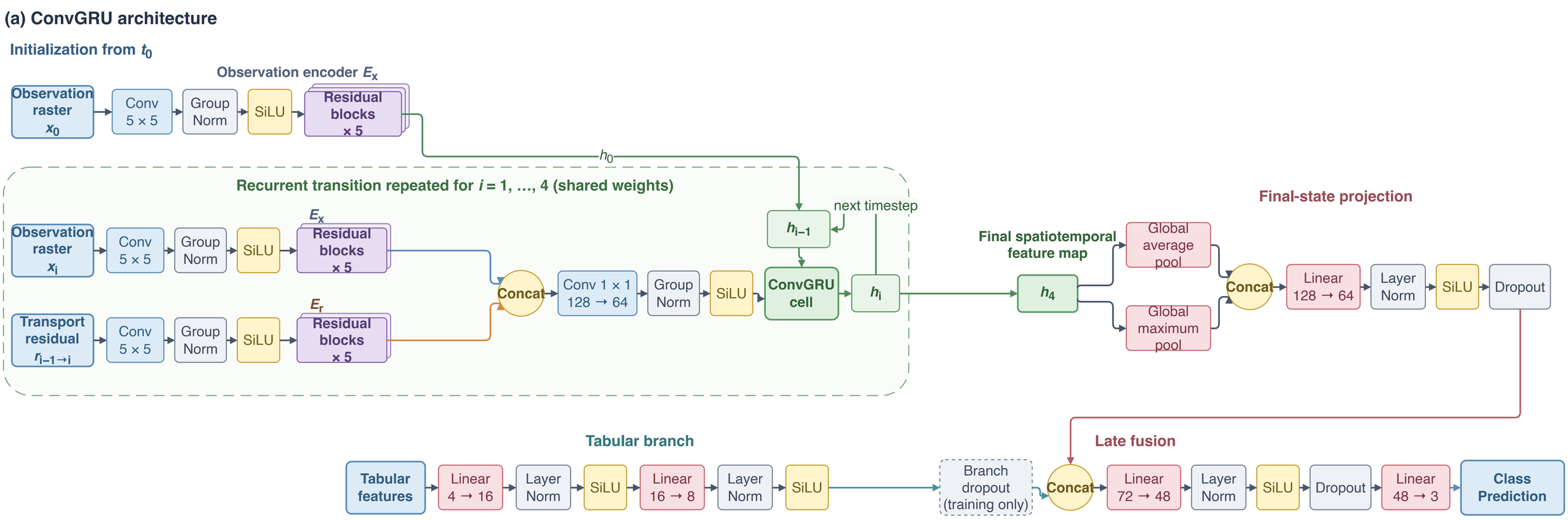

(a) ConvGRU architecture
Initialization from t_0
Observation encoder E_x
Observation raster x_0
Conv 5 × 5
Group Norm
SiLU
Residual blocks × 5
h_0
Recurrent transition repeated for i = 1, …, 4 (shared weights)
Observation raster x_i
Conv 5 × 5
Group Norm
SiLU
E_x
Residual blocks × 5
Transport residual $r_{i-1→i}$
Conv 5 × 5
Group Norm
SiLU
E_r
Residual blocks × 5
Concat
Conv 1 × 1 128 → 64
Group Norm
SiLU
ConvGRU cell
h_{i-1}
next timestep
h_i
Final-state projection
Final spatiotemporal feature map
h_4
Global average pool
Global maximum pool
Concat
Linear 128 → 64
Layer Norm
SiLU
Dropout
Tabular branch
Tabular features
Linear 4 → 16
Layer Norm
SiLU
Linear 16 → 8
Layer Norm
SiLU
Branch dropout (training only)
Late fusion
Concat
Linear 72 → 48
Layer Norm
SiLU
Dropout
Linear 48 → 3
Class Prediction